

Reachable Nodes In Subdivided Graph (LC 882)

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Core Concepts & Observations

Edge Information: (u, v, cnt) gives no. of new nodes to be inserted in edge connecting (u, v) .

Example: suppose edge $\rightarrow 0 \text{ ---- } 1$, $cnt = 4$
then $0 - a - b - c - d - 1$ and $maxmoves = 2$
then $0 - a - b \rightarrow$ reachable
(the arrows indicate no of reachable nodes)

Cant use BFS as edges have weights, $(cnt + 1)$ edges

Algorithm Steps

- S1** make adjacency list
- S2** Run Dijkstras to know the minimum moves required to reach every **original** node
- S3** count original nodes $\rightarrow$ every node whose distance $\leq maxmoves$
- S4** count subnodes $\rightarrow \min(cnt, \text{left from } u + \text{left from } v)$

Time Complexity (TC): $O(E) + O(E \log V) + O(E)$ (build + Dijkstra + process edges)

Implementation

```
class Solution {
public:
    int reachableNodes(vector<vector<int>>& edges, int maxmoves, int n) {
```

```

vector<vector<pair<int,int>>> adj(n);
for(auto it:edges){

    int u=it[0];
    int v=it[1];
    int cnt=it[2];

    adj[u].push_back({v,cnt});
    adj[v].push_back({u,cnt});

}

//we need minimum cost to reach each node to save on moves

priority_queue<pair<int,int>,vector<pair<int,int>>,greater<pair<int,int>>>
pq;

vector<int> mindist(n,INT_MAX);

pq.push({0,0});
mindist[0]=0;

while(!pq.empty()){
    int currnode=pq.top().second;
    int currdis=pq.top().first;
    pq.pop();

    if(currdis>mindist[currnode])
        continue;

    for(auto &it:adj[currnode]){
        int newnode=it.first;
        int cost=it.second;
        int newdis=currdis+cost+1;

        if(newdis<mindist[newnode]){
            mindist[newnode]=newdis;
            pq.push({newdis,newnode});
        }

    }
}

int ans=0;
for(int dis:mindist){
    if(dis<=maxmoves){
        ans++;
    }
}

```

```
for(auto &it:edges){
    int u=it[0];
    int v=it[1];
    int cnt=it[2];

    int uwaale=max(0,maxmoves-mindist[u]); //pohovhne ke baad bhi thoda
sa bach gya ,usse subnode jaayenge
    int vwaale=max(0,maxmoves-mindist[v]);

    ans+=min(cnt,uwaale+vwaale);
}

return ans;
}
};
```